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import nltk
import string
import matplotlib.pyplot as plt
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.probability import FreqDist

nltk.download('punkt')
nltk.download('stopwords')

articles = [
    "Artificial intelligence is transforming healthcare and education systems",
    "AI and machine learning are the future of technology and innovation",
    "Healthcare systems are improving with artificial intelligence solutions",
    "Education technology is growing rapidly with AI based tools"
]

text = " ".join(articles).lower()

tokens = word_tokenize(text)

stop_words = set(stopwords.words('english'))

filtered_tokens = [
    word for word in tokens
    if word not in stop_words
    and word not in string.punctuation
]

print("Tokens:")
print(filtered_tokens)

freq_dist = FreqDist(filtered_tokens)

print("\nWord Frequency Distribution:")
for word, freq in freq_dist.items():
    print(word, ":", freq)

vocabulary = set(filtered_tokens)

print("\nVocabulary:")
print(vocabulary)

print("\nVocabulary Size:", len(vocabulary))

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top5 = freq_dist.most_common(5)

print("\nTop 5 Frequent Words:")
for word, count in top5:
    print(word, ":", count)

total_tokens = len(filtered_tokens)
unique_tokens = len(vocabulary)

ttr = unique_tokens / total_tokens

print("\nType-Token Ratio (TTR):", round(ttr, 3))

topics = {
    "AI": ["ai", "artificial", "intelligence", "machine", "learning"],
    "Healthcare": ["healthcare", "systems"],
    "Education": ["education"],
    "Technology": ["technology", "innovation", "tools"]
}

topic_scores = {}

for topic, keywords in topics.items():
    score = sum(freq_dist[word] for word in keywords)
    topic_scores[topic] = score

dominant_topic = max(topic_scores, key=topic_scores.get)

print("\nTopic Scores:")
for topic, score in topic_scores.items():
    print(topic, ":", score)

print("\nDominant Topic:", dominant_topic)
words = [word for word, freq in top5]
frequencies = [freq for word, freq in top5]

plt.figure(figsize=(8,5))
plt.bar(words, frequencies)
plt.title("Top 5 Frequent Words")
plt.xlabel("Words")
plt.ylabel("Frequency")
plt.show()

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